Test Series No.

02

NEET Success Kit [2026-27]

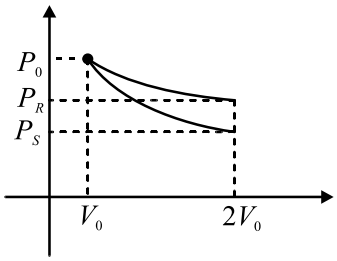
DURATION: 195 Minutes | **M. MARKS: 720**

DATE: 10/06/2026

ANSWER KEY

1.	(2)	46.	(2)	91.	(4)	136.	(4)
2.	(1)	47.	(2)	92.	(3)	137.	(1)
3.	(2)	48.	(1)	93.	(2)	138.	(2)
4.	(2)	49.	(3)	94.	(2)	139.	(1)
5.	(4)	50.	(2)	95.	(1)	140.	(3)
6.	(2)	51.	(1)	96.	(1)	141.	(2)
7.	(2)	52.	(4)	97.	(4)	142.	(2)
8.	(2)	53.	(4)	98.	(4)	143.	(4)
9.	(3)	54.	(2)	99.	(2)	144.	(2)
10.	(4)	55.	(2)	100.	(1)	145.	(1)
11.	(4)	56.	(1)	101.	(2)	146.	(3)
12.	(3)	57.	(3)	102.	(3)	147.	(3)
13.	(3)	58.	(4)	103.	(4)	148.	(1)
14.	(1)	59.	(1)	104.	(4)	149.	(4)
15.	(4)	60.	(4)	105.	(2)	150.	(1)
16.	(4)	61.	(2)	106.	(1)	151.	(3)
17.	(1)	62.	(3)	107.	(3)	152.	(3)
18.	(2)	63.	(2)	108.	(1)	153.	(4)
19.	(2)	64.	(2)	109.	(4)	154.	(1)
20.	(2)	65.	(2)	110.	(3)	155.	(3)
21.	(2)	66.	(1)	111.	(1)	156.	(3)
22.	(1)	67.	(1)	112.	(4)	157.	(3)
23.	(2)	68.	(1)	113.	(3)	158.	(4)
24.	(3)	69.	(3)	114.	(3)	159.	(2)
25.	(4)	70.	(4)	115.	(3)	160.	(1)
26.	(4)	71.	(1)	116.	(1)	161.	(3)
27.	(2)	72.	(3)	117.	(3)	162.	(4)
28.	(4)	73.	(4)	118.	(3)	163.	(3)
29.	(3)	74.	(2)	119.	(2)	164.	(2)
30.	(2)	75.	(1)	120.	(1)	165.	(3)
31.	(3)	76.	(1)	121.	(2)	166.	(2)
32.	(1)	77.	(1)	122.	(2)	167.	(2)
33.	(3)	78.	(3)	123.	(3)	168.	(4)
34.	(4)	79.	(2)	124.	(2)	169.	(2)
35.	(1)	80.	(2)	125.	(1)	170.	(1)
36.	(1)	81.	(3)	126.	(3)	171.	(3)
37.	(3)	82.	(1)	127.	(3)	172.	(2)
38.	(1)	83.	(1)	128.	(2)	173.	(3)
39.	(3)	84.	(1)	129.	(1)	174.	(3)
40.	(1)	85.	(3)	130.	(3)	175.	(3)
41.	(3)	86.	(1)	131.	(4)	176.	(3)
42.	(3)	87.	(1)	132.	(2)	177.	(4)
43.	(4)	88.	(1)	133.	(3)	178.	(3)
44.	(3)	89.	(2)	134.	(2)	179.	(2)
45.	(1)	90.	(1)	135.	(2)	180.	(3)

1. (2)



$P_0V_0 = P_R(2V_0) \Rightarrow P_R = \frac{P_0}{2}$

$P_0V_0^{\gamma} = P_S(2V_0)^{\gamma} \Rightarrow P_S = \frac{P_0}{2^{\gamma}}$

$\therefore \frac{P_R}{P_S} = \frac{P_0(2^{\gamma})}{2(P_0)} = 2^{\gamma-1}$

(NCERT Ed. 2025-26, 11th, Page No. 235)

2. (1)

$R = \frac{V^2}{P} = \frac{220 \times 220}{250}$

Power when connected to a 110V dc supply

$P = \frac{110 \times 110 \times 250}{220 \times 220} = \frac{250}{4} = 62.5 \text{ W}$

(NCERT Ed. 2025-26, 12th, Page No. 92)

3. (2)

By the Equation of Continuity,

$A_1v_1 = A_2v_2 \Rightarrow v \propto \frac{1}{A} \propto \frac{1}{d^2}$.

Minimum velocity occurs at the maximum diameter, and maximum velocity at the minimum diameter.

$\frac{v_{min}}{v_{max}} = \left(\frac{d_{min}}{d_{max}}\right)^2 = \left(\frac{3.0}{7.5}\right)^2 = \left(\frac{2}{5}\right)^2 = \frac{4}{25}$.

(NCERT Ed. 2025-26, 11th, Page No. 187)

4. (2)

For a cylindrical source $I \propto \frac{1}{r}$

(NCERT Ed. 2025-26, 12th, Page No. 34)

5. (4)

Energy density

$= \frac{1}{2}\varepsilon_0 E^2 = \frac{1}{2}\varepsilon_0 \frac{V^2}{d^2} = \frac{1}{2}\varepsilon_0 \frac{V^2C^2}{\varepsilon_0^2 A^2} = \frac{V^2C^2}{2\varepsilon_0 A^2}$

Energy = Energy density × volume $= \frac{C^2V^2a^3}{2\varepsilon_0 A^2}$

(NCERT Ed. 2025-26, 12th, Page No. 75)

6. (2)

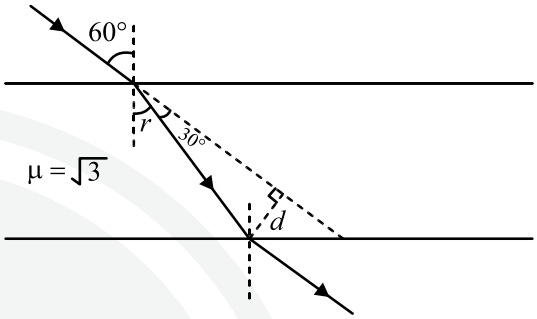
$a = \frac{50-30}{8} = \frac{20}{8} = 2.5 \text{ m/s}^2$

$v_A = 2.5 \times 2 = 5.0 \text{ m/s}$

Speed of image of block A is equal to v_A

(NCERT Ed. 2025-26, 12th, Page No. 222)

7. (2)



$1.\sin 60° = \sqrt{3}\sin r$

$\Rightarrow \frac{\sqrt{3}}{2} = \sqrt{3}\sin r \Rightarrow r = 30°$

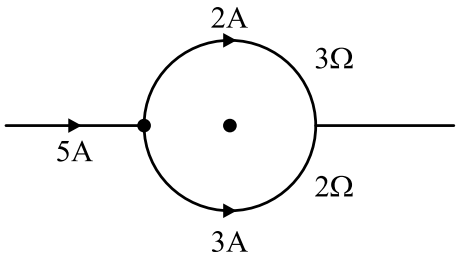
$d = t \sin 30°$

$= \frac{t}{2} = \frac{4}{2} = 2 \text{ cm}$

(NCERT Ed. 2025-26, 12th, Page No. 229)

8. (2)

Distribution of current



The magnetic field due to both will be in opposite directions

$B_{net} = B_1 - B_2 = \frac{\mu_0 3}{4(1)} - \frac{\mu_0(2)}{4(1)} = \frac{\mu_0}{4} = \frac{4\pi \times 10^{-7}}{4}$

$= \pi \times 10^{-7} \text{ T}$

(NCERT Ed. 2025-26, 12th, Page No. 116)

[Contd...

9. (3)

Let the temperature at distance x from end A is T.

Equating the heat currents:

$$\frac{T_H - T}{x} = \frac{T - T_L}{L - x}$$

$$\Rightarrow T_H(L-x) - T(L-x) = Tx - T_L x$$

$$\Rightarrow T_H L - T_H x - TL + Tx = Tx - T_L x$$

$$\Rightarrow T_H L - TL = (T_H - T_L)x$$

$$\Rightarrow T = T_H - (T_H - T_L)\frac{x}{L}$$

Represents a straight line with negative slope and positive Y intercept.

(NCERT Ed. 2025-26, 11th, Page No. 215)

10. (4)

Conservation of mechanical energy loss in gravitational potential energy = Gain in kinetic energy

$$\Rightarrow mg\frac{L}{2}\sin 37° = \frac{1}{2} \times \frac{mL^2}{3} \times \omega^2$$

(NCERT Ed. 2025-26, 11th, Page No. 115)

11. (4)

0.205 has least number of digits after decimal and it is equal to 3 which is obtained in 0.004.

(NCERT Ed. 2025-26, 11th, Page No. 5)

12. (3)

At resonance the impedance is minimum, therefore the current in the circuit is maximum.

Power factor at resonance is 1.

The circuit becomes purely resistive, so voltage and current are in phase.

(NCERT Ed. 2025-26, 12th, Page No. 189)

13. (3)

$$v_{max} = \omega A$$

$$v_{max} = \frac{\omega A}{2} = \omega\sqrt{A^2 - x^2}$$

$$\Rightarrow \frac{A^2}{4} = A^2 - x^2$$

$$\Rightarrow x^2 = \frac{3A^2}{4}$$

$$\therefore x = \frac{\sqrt{3}A}{2}$$

(NCERT Ed. 2025-26, 11th, Page No. 266)

14. (1)

Energy released (Q) in a nuclear reaction is the difference between the total binding energy of the products and the total binding energy of the reactants.

Reaction: $3B \rightarrow 1A + Q$

$$Q = (\text{B.E. of Product}) - (\text{B.E. of Reactants})$$

$$= E_a - 3E_b .$$

(NCERT Ed. 2025-26, 12th, Page No. 312)

15. (4)

Circumference $S = 2\pi R = 2\pi(0.5) = \pi$ m

Relative speed $v_{rel} = 25\pi - 20\pi = 5\pi$ cm/s

Time to meet $t = \frac{S}{v_{rel}} = \frac{100\pi}{5\pi} = 20$ s

(NCERT Ed. 2025-26, 11th, Page No. 24)

16. (4)

$$\frac{11v}{4l_c} = \frac{4v}{2l_o} \Rightarrow \frac{l_o}{l_c} = \frac{8}{11}$$

(NCERT Ed. 2025-26, 11th, Page No. 291)

17. (1)

Newton's law of cooling.

Rate of loss of heat $\propto$ difference in temperature

$$\frac{30}{6} = K \times \left(\frac{80+50}{2} - 20\right) \quad ...(i)$$

$$\frac{20}{t} = K \times \left(\frac{60+40}{2} - 20\right) \quad ...(ii)$$

Divide (i) by (ii)

$$\frac{5t}{20} = 1.5 \Rightarrow t = 6 \text{ minutes}$$

(NCERT Ed. 2025-26, 11th, Page No. 221)

18. (2)

According to KTG,

The force of interaction among gas molecules is negligible.

(NCERT Ed. 2025-26, 11th, Page No. 248)

[Contd...

19. (2)

Gate I (AND): $0 \cdot 1 = 0$

Gate II (NOR): $\overline{0+1} = \overline{1} = 0$

Gate III (NAND): $\overline{1 \cdot 1} = \overline{1} = 0$

Gate IV (OR): $0+1=1$

Only Gate IV has an output of 1; Gates I, II, and III have an output of 0.

(NCERT Ed. 2025-26, 12th, Page No. 341)

20. (2)

With height $g' = \dfrac{gR^2}{(R+H)^2}$

When $H = R$ $\quad g' = \dfrac{g}{4}$

When $H = 2R$ $\quad g' = \dfrac{g}{9}$

With depth $g' = g\left(1 - \dfrac{d}{R}\right)$

When $d = R$ $\quad g' = 0$

When $d = \dfrac{R}{2}$ $\quad g' = \dfrac{g}{2}$

(NCERT Ed. 2025-26, 11th, Page No. 134)

21. (2)

$$R = \frac{\rho l}{A}$$

$$\frac{R_{\max}}{R_{\min}} = \frac{5 \times 5 \times 2}{1 \times 2 \times 1} = \frac{25}{1}$$

(NCERT Ed. 2025-26, 12th, Page No. 84)

22. (1)

$$\frac{4T}{R_1} = 3\left(\frac{4T}{R_2}\right) \Rightarrow R_2 = 3R_1$$

Ratio of surface area $= \dfrac{R_1^2}{R_2^2} = \dfrac{1}{9}$

(NCERT Ed. 2025-26, 11th, Page No. 196)

23. (2)

$$W = \int_4^{-2} F dx = -\int_4^{-2} 6x^3 dx = \frac{-3}{2}\left[2^4 - 4^4\right]$$

$$= \frac{3}{2} \times 240 = 360\,\text{J}$$

(NCERT Ed. 2025-26, 11th, Page No. 75)

24. (3)

When ice converts into 0°C water heat required

= 100 × 80 = 8000 cal

Heat lost by water to reach 0°C

= 100 × 80 × 1 = 8000 cal

Heat lost = Heat gained

$\therefore$ Final temperature is 0°C

(NCERT Ed. 2025-26, 11th, Page No. 213)

25. (4)

Bulk modulus is defined for all the three states

Shear modulus and Young's modulus are defined only for solids.

(NCERT Ed. 2025-26, 11th, Page No. 173)

26. (4)

$$K = \frac{1}{2} I \omega^2$$

$$I = \frac{ML^2}{12} + M\left(\frac{L}{4}\right)^2 = \frac{7ML^2}{48}$$

$$K = \frac{1}{2} \times \frac{7}{48} \times 2 \times 1^2 \times 144 = 21\,\text{J}$$

(NCERT Ed. 2025-26, 11th, Page No. 119)

27. (2)

$$\Delta Q = \frac{\Delta \phi}{R}$$

$$\phi(2) = 5(2)^2 - 4(2) + 1$$

$$\phi(2) = (20 - 8 + 1)\,\text{Wb} = 13\ \text{Wb}$$

$$\phi(0) = 1\,\text{Wb}$$

$$\Delta Q = \frac{(13-1)}{10}\ \text{Wb} = 1.2\ \text{C}$$

(NCERT Ed. 2025-26, 12th, Page No. 158)

28. (4)

$$P_{rms} = V_{rms} . I_{rms} \cos\phi$$

$$P_{rms} = V_{rms} . I_{rms} \frac{R}{Z} = \frac{V_{rms} I_{rms} R}{Z}$$

$$P_{rms} = I_{rms}^2 R$$

$$R = \frac{80}{16} = 5\Omega$$

$$Z = \frac{V}{I} = 50\Omega$$

$$5^2 + X_L^2 = (50)^2$$

$$X_L^2 = (50)^2 - 5^2$$

$$X_L = \sqrt{(55)(45)}$$

$$X_L = 15\sqrt{11}\Omega$$

(NCERT Ed. 2025-26, 12th, Page No. 187)

[Contd...

29. (3)

$$T \propto \frac{n^3}{z^2}$$

$$\frac{T_2}{T_3} = \left(\frac{n_2}{n_3}\right)^3$$

$$\frac{12.8\times10^{-16}}{T_3} = \left(\frac{2}{3}\right)^3$$

$$T_3 = \frac{27\times12.8\times10^{-16}}{8}$$

$$f_3 = \frac{8}{27\times12.8\times10^{-16}}$$

$$= \frac{8}{345.6}\times10^{16}\ s^{-1} = 2.3 \times 10^{14}\ s^{-1}$$

(NCERT Ed. 2025-26, 12th, Page No. 300)

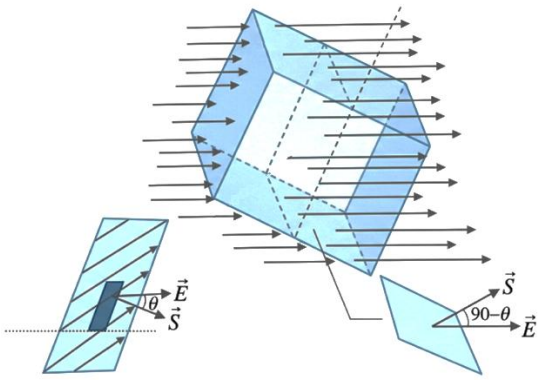
30. (2)

The magnitude of net force acting on man

$= 45 \times 2 = 90$ N

(NCERT Ed. 2025-26, 11th, Page No. 54)

31. (3)



total flux of electric field entering the cube in this position = flux through front tilted face + flux through bottom tilted face

$\phi_{total} = \phi_1 + \phi_2$

$= EL^2 (\cos\theta + \sin\theta)$

Express $(\cos\theta + \sin\theta)$ as a square root to match the choices:

$$\cos\theta + \sin\theta = \sqrt{(\cos\theta + \sin\theta)^2}$$

$$= \sqrt{\cos^2\theta + \sin^2\theta + 2\sin\theta\cos\theta}$$

$$= \sqrt{1 + \sin(2\theta)}$$

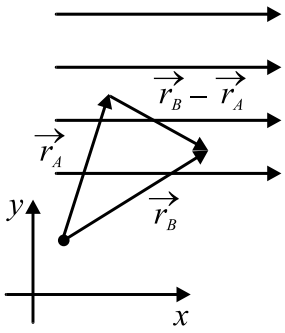
Therefore, the flux is $E \cdot l^2 \sqrt{1 + \sin(2\theta)}$

(NCERT Ed. 2025-26, 12th, Page No. 31)

32. (1)

$\vec{r}_A = 2\hat{i} + 2\hat{j}$

$\vec{r}_B = 3\hat{i} + 4\hat{j}$



$\vec{r}_B - \vec{r}_A = \hat{i} + 2\hat{j}$

$(\vec{r}_B - \vec{r}_A).(\vec{E}) = (\hat{i} + 2\hat{j}).(E_0(\hat{i} + \hat{j})) = 3E_0$ which is positive.

Therefore A is at higher potential.

(NCERT Ed. 2025-26, 12th, Page No. 55)

33. (3)

$$F = \frac{kQ^2}{r^2}$$

$$F_1 = \frac{K\left(\frac{4Q}{5}\right)^2}{r^2} = \frac{16F}{25}$$

(NCERT Ed. 2025-26, 12th, Page No. 8)

34. (4)

Using $h = \frac{2T\cos\theta}{r\rho g} \Rightarrow T \propto \frac{h\rho}{\cos\theta}$

$$\frac{T_w}{T_L} = \frac{h_w\rho_w\cos\theta_L}{h_L\rho_L\cos\theta_w} = \frac{6\times1\times\left|\cos120^\circ\right|}{2\times9\times\cos0^\circ}$$

$$= \frac{6\times0.5}{18\times1} = \frac{3}{18} = 1:6$$

(NCERT Ed. 2025-26, 11th, Page No. 197)

35. (1)

Percentage change in length due to temperature change

$$\%\Delta l = \frac{\Delta l}{l}\times100 = \alpha\Delta T\times100$$

$$= 2\times10^{-5}\times100\times100 = 0.2\%$$

(NCERT Ed. 2025-26, 11th, Page No. 205)

[Contd...

36. (1)

$P_{eq} = +5D$

$f_{eq} = \frac{1}{P_{eq}} = \frac{100}{5}\text{cm}$

$f_{eq} = 20\text{cm}$

Magnifying power $= \frac{D}{f_{eq}} = \frac{25}{20} = 1.25$

(NCERT Ed. 2025-26, 12th, Page No. 242)

37. (3)

Width of central maxima $= \frac{2\lambda D}{a}$

$= \frac{2\times500\times10^{-9}\times2}{10^{-3}} = 2\times10^{-3}\text{ m} = 2\text{ mm}$

(NCERT Ed. 2025-26, 12th, Page No. 267)

38. (1)

Pressure = $[ML^{-1}T^{-2}]$

Velocity = $[M^0L^1T^{-1}]$

Acceleration = $[M^0L^1T^{-2}]$

Force = $[M^1L^1T^{-2}]$

(NCERT Ed. 2025-26, 11th, Page No. 7)

39. (3)

The equation of trajectory of projectile is

$y = x\left(\frac{3}{4}\right) - \frac{10x^2\times25}{2\times400\times16}$

$y = \frac{3x}{4} - \frac{5x^2}{256}$

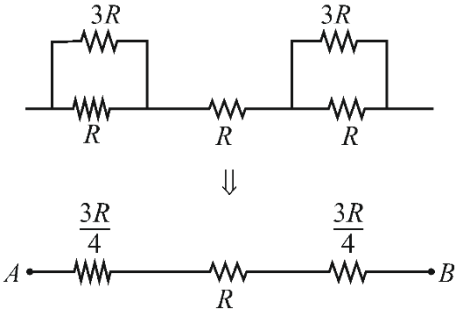
Point $(R, -R)$ satisfies the equation

$-R = \frac{3R}{4} - \frac{5R^2}{256}$

$\Rightarrow \frac{-7R}{4} = -\frac{5R^2}{256} \Rightarrow R = \frac{448}{5}$ m.

(NCERT Ed. 2025-26, 11th, Page No. 40)

40. (1)



$R_{eq} = \frac{3R}{4} + \frac{3R}{4} + R = \frac{5R}{2}$

(NCERT Ed. 2025-26, 12th, Page No. 102)

41. (3)

$P = \frac{E}{c}$

$P = \frac{9\times10^6}{3\times10^8}\text{ eV s m}^{-1}$

$P = 0.03\text{ eV s m}^{-1}$

(NCERT Ed. 2025-26, 12th, Page No. 283)

42. (3)

$\frac{1}{f_{eq}} = \frac{1}{f_1} + \frac{1}{f_2}$

$\frac{1}{f_{eq}} = \frac{1}{20} + \frac{1}{40}$

$\frac{1}{f_{eq}} = \frac{3}{40} \Rightarrow f_{eq} = \frac{40}{3}$ cm

(NCERT Ed. 2025-26, 12th, Page No. 238)

43. (4)

$q = CV$

$\frac{dq}{dt} = C\frac{dV}{dt}$

$I_D = 5\times10^{-6}\times10^7 = 50\text{ A}$

(NCERT Ed. 2025-26, 12th, Page No. 203)

44. (3)

$S_1 : S_2 : S_3 : S_4 = 1 : 3 : 5 : 7$

$\Rightarrow \frac{S_1 + S_2}{S_3 + S_4} = \frac{1+3}{5+7} = \frac{1}{3}$

(NCERT Ed. 2025-26, 11th, Page No. 20)

45. (1)

Both A and R are false

Kinetic frictional force can do positive work. Kinetic frictional force can also act in the direction of velocity for example in the case of a two block system in which there is slipping between the blocks.

The upper block is accelerated due to kinetic friction.

(NCERT Ed. 2025-26, 11th, Page No. 62)

46. (2)

$$CH_3-CH=CH-\overset{*}{\underset{\displaystyle CH_3}{\underset{|}{C}}}H-CH=CH_2$$

Total stereogenic site = 2

$\therefore$ total stereoisomers = $2^2 = 4$.

(NEW NCERT 11th Part-II Page No. 308)

47. (2)

Tin and lead are soft metals with low melting points.

(OLD NCERT 11th Part-I Page No. 324)

48. (1)

- Enantiomers have same melting point.
- Acetone is polar aprotic solvent.

(NEW NCERT 12th Part-II Page No. 177, 180)

49. (3)

$$N_2 + 3H_2 \longrightarrow 2NH_3$$

(i) 4 mol 9 mol

(f) $\left(4-\frac{9}{3}\right)$mol – $\left(\frac{2}{3}\times 9\right)$mol

$= 6$ mol

(NEW NCERT 11th Part-I Page No. 20)

50. (2)

'spin only' magnetic moment, $\mu = \sqrt{n(n+2)}$ BM

- $Cr^{3+}(Z=24):[Ar]3d^3 4s^0$

$\mu = \sqrt{3(3+2)} = \sqrt{15}$ BM

- $Fe^{3+}(Z=26):[Ar]3d^5 4s^0$

$\mu = \sqrt{5(5+2)} = \sqrt{35}$ BM

- $Sc^{3+}(Z=21):[Ar]3d^0 4s^0$

$\mu = \sqrt{0(0+2)} = 0$ BM

- $Ti^{3+}(Z=22):[Ar]3d^1 4s^0$

$\mu = \sqrt{1(1+2)} = \sqrt{3}$ BM

(NEW NCERT 12th Part-I Page No. 102)

51. (1)

- True. In acidic medium, $MnO_4^- \rightarrow Mn^{2+}$.
- Mn goes from +7 to +2, gain = $5e^-$. So 1 mole permanganate accepts 5 moles electrons.
- True. Each $Fe^{2+} \rightarrow Fe^{3+}$ loses 1 electron, so 5 moles Fe^{2+} provide 5 electrons.
- False. +7 to +4 occurs in neutral/basic medium (to MnO_2), not acidic.

(NCERT Ed. 2025-26 11th Part-II Page No. 239)

52. (4)

$$\bar{\nu} = \frac{1}{\lambda} = RZ^2\left(\frac{1}{n_1^2}-\frac{1}{n_2^2}\right)$$

For shortest λ in Balmer series,

$n_1 = 2$ and $n_2 = \infty$

$$\bar{\nu} = R\times 1^2\times\left(\frac{1}{2^2}-\frac{1}{\infty^2}\right) = \frac{R}{4}$$

(NEW NCERT 11th Part-I Page No. 48)

53. (4)

Born-Haber cycle is used as indirect method to determine lattice enthalpies of ionic compounds.

(NEW NCERT 11th Part-I Page No. 140, 157, 159, 155)

54. (2)

- $C_2H_5Cl \xrightarrow[^-CN]{KCN} C_2H_5CN$
- $C_2H_5Cl \xrightarrow{AgC\ddot{N}} C_2H_5NC$
- $C_2H_5Cl \xrightarrow{Ag\ddot{O}NO} C_2H_5NO_2$
- $C_2H_5Cl \xrightarrow{\overset{+}{K}\overset{-}{O}NO} C_2H_5ONO$

(NEW NCERT 12th Part-II Page No. 170)

55. (2)

$$\overset{0}{Cl_2} + 2\overset{-1}{K}I \longrightarrow 2K\overset{-1}{Cl} + \overset{0}{I_2}$$

Reduction (Cl: 0 → −1); Oxidation (I: −1 → 0)

Redox reactions involve change in oxidation number of the interacting species.

(NEW NCERT 11th Part-II Page No. 241)

56. (1)

For first order: $k = \frac{0.693}{t_{1/2}} = \frac{0.693}{5} = 0.1386\ min^{-1}$

$$\frac{1}{8} = \left(\frac{1}{2}\right)^3 \Rightarrow t = 3t_{1/2} = 15\ min$$

[Contd...

75% completion $\Rightarrow$ 25% remaining $= \frac{1}{4} = \left(\frac{1}{2}\right)^2$

$\Rightarrow t = 2t_{1/2} = 10 \text{ min}$

Half-life of first order reaction is independent of initial concentration

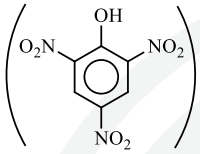
(NCERT Ed. 2025-26 12th Part-I Page No. 71)

57. (3)

$[CrCl_3(NH_3)_3]$: Triamminetrichloridochromium(III)

(NEW NCERT 12th Part-I Page No. 123)

58. (4)

- Phenol does not react with aq. $NaHCO_3$
- Picric acid  is more acidic than phenol

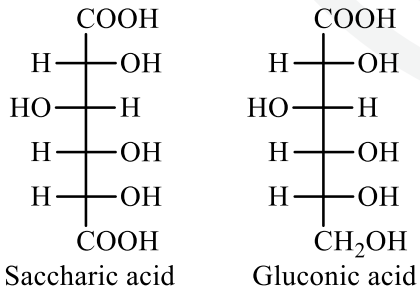
(NEW NCERT 12th Part-II Page No. 207)

59. (1)

O_2 molecule is paramagnetic in nature as it contains two unpaired electrons in π^*2p_x and π^*2p_y molecular orbitals.

(NEW NCERT 11th Part-I Page No. 130)

60. (4)



(NEW NCERT 12th Part-II Page No. 284)

61. (2)

$KMnO_4$ acts as self indicator in the titration of $KMnO_4$ vs $H_2C_2O_4$. The end point is indicated by the appearance of a light pink colour

(NEW NCERT 11th Part-II Page No. 249)

62. (3)

$$CH_3CH_2OH \xrightarrow{PCl_5} CH_3CH_2Cl \xrightarrow{KCN}$$

$$CH_3CH_2CN \xrightarrow{H_3O^+} CH_3CH_2COOH$$

(NEW NCERT 12th Part-II Page No. 246)

63. (2)

Add the two given reactions:

$A + B \rightarrow C$

$C + B \rightarrow D$

On adding: $A + 2B \rightarrow D$

When reactions are added, equilibrium constants multiply.

$K = K_1 \times K_2$

(NCERT Ed. 2025-26 11th Part-I Page No. 182)

64. (2)

$n_{solute} = 0.2$

$n_{total} = 1$

$N_{H_2O} = 1 - 0.2 = 0.8$

$W_{H_2O} = 0.8 \times 18 \text{ g}$

$$\text{Molality(m)} = \frac{0.2}{0.8 \times 18 \times 10^{-3}} = 13.88 \text{ m}$$

(NEW NCERT 11th Part-I Page No. 24)

65. (2)

For H-atom energy of orbitals:

$1s < 2s = 2p < 3s = 3p = 3d < 4s$

(NEW NCERT 11th Part-I Page No. 60)

66. (1)

$\Delta_{eg}H$ generally becomes more negative across a period as we move left to right. Within a group, $\Delta_{eg}H$ becomes less negative down a group. However, adding an electron to the 2p-orbital leads to greater repulsion than adding an electron to the larger 3p-orbital.

(NEW NCERT 11th Part-I Page No. 90)

67. (1)

- True (BF_3 trigonal planar, sp^2)
- True (NH_4^+ has 4 σ bonds, no lone pair)
- True (CO has bond order 3; all electrons paired $\rightarrow$ diamagnetic)

[Contd...

- O_2 has two unpaired electron, so it is paramagnetic, and bond order is 2

(NCERT Ed. 2025-26 11th Part-I Page No. 113)

68. (1)

$$\Delta S_{surr} = -\frac{\Delta_f H^o}{T} = -\frac{(-286\times10^3)}{298} = 959.73\ JK^{-1}$$

(NEW NCERT 11th Part-I Page No. 167)

69. (3)

$$2CH_3COOH \underset{\text{Benzene}}{\rightleftharpoons} (CH_3COOH)_2$$

$$i = \frac{1}{2} = 0.5$$

(NEW NCERT 12th Part-I Page No. 24)

70. (4)

Cr in CrO_4^{2-} is present in its maximum oxidation state +6 so it cannot disproportionate.

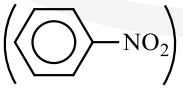
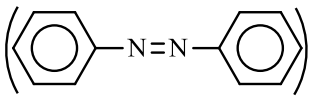
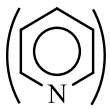
(NEW NCERT 11th Part-II Page No. 244)

71. (1)

- $NaNO_3$: Chile Saltpetre
- $MgSO_4\cdot7H_2O$: Epsom salt
- PbS: Galena
- Na_3AlF_6: Cryolite

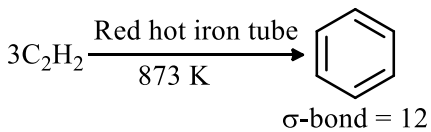
(OLD NCERT 12th Part-I Page No. 165, 180, 192)

72. (3)

Kjeldahl method is not applicable to compounds containing nitrogen in nitro groups 
and azo groups  and nitrogen present in the ring  .

(NEW NCERT 11th Part-II Page No. 288)

73. (4)



σ-bond = 12

(NEW NCERT 11th Part-II Page No. 318)

74. (2)

$$Ca(OH)_2 \rightarrow \underset{0.1\,M}{Ca^{2+}} + \underset{0.2\,M}{2OH^-}$$

$$Ni(OH)_2 \rightleftharpoons \underset{S}{Ni^{2+}} + \underset{2S}{2OH^-}$$

$$K_{sp} = [Ni^{2+}][OH^-]^2$$

$$2\times10^{-15} = (S)\,(2s+0.2)^2$$

$$2\times10^{-15} \simeq S\times(0.2)^2$$

$$S = 5\times10^{-14}\ M$$

(NEW NCERT 11th Part-I Page No. 206)

75. (1)

For dilute non-electrolytes: π = CRT

At same T, $\pi \propto C = \frac{\text{moles}}{\text{volume}}$.

All volumes are equal (300 mL), so compare moles only.

Moles in each:

- p_1 : glucose moles $= \frac{12}{180} = 0.0667$
- p_2 : urea moles $= \frac{6}{60} = 0.10$
- p_3 : sucrose moles $= \frac{17.1}{342} = 0.05$

So, $p_2(0.10) > p_1(0.0667) > p_3(0.05)$

(NCERT Ed. 2025-26 12th Part-I Page No. 10)

76. (1)

$$Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$$

When opposite $E_{ext} > 1.1$ V

- Electrons flow from Cu to Zn.
- Current flows from Zn to Cu.
- Zn is deposited at the Zn electrode.

(NEW NCERT 12th Part-I Page No. 32)

77. (1)

- $\overset{+6}{K_2Cr_2O_7} \xrightarrow{2\times3e^-} 2Cr^{3+}$
- $\overset{0}{I_2} \xrightarrow{2\times5e^-} 2\overset{+5}{IO_3^-}$
- $Fe^{2+} \xrightarrow{1e^-} Fe^{3+}$
- $\overset{+2}{NO} \xrightarrow{3e^-} \overset{+5}{NO_3^-}$

(NEW NCERT 12th Part-I Page No. 52)

[Contd...

78. (3)

Inversion of cane sugar is pseudo first order reaction.

$$\underset{\text{cane sugar}}{C_{12}H_{22}O_{11}} + H_2O \xrightarrow{H^+} \underset{\text{glucose}}{C_6H_{12}O_6} + \underset{\text{fructose}}{C_6H_{12}O_6}$$

$$\text{rate} = k[C_{12}H_{22}O_{11}]$$

(NEW NCERT 12th Part-I Page No. 78, 73, 72)

79. (2)

$\Delta_a H^o : V > Ti > Cr > Mn$

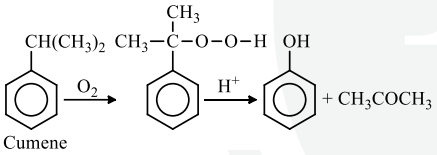
(NEW NCERT 12th Part-I Page No. 94)

80. (2)

Mn in MnO_4^- is +7; in MnO_4^{2-} is +6. Both statements are true, but "higher oxidation state" alone isn't a complete explanation of colour intensity (colour arises from electronic transitions/charge transfer).

(NCERT Ed. 2025-26 12th Part-I Page No. 113)

81. (3)



(NEW NCERT 12th Part-II Page No. 202)

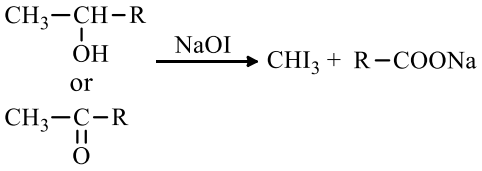
82. (1)

$CHCl_3$ was used as general anaesthetic.

(NEW NCERT 12th Part-II Page No. 187)

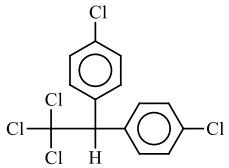
83. (1)

Iodoform reaction:



(NEW NCERT 12th Part-II Page No. 240)

84. (1)

DDT: 

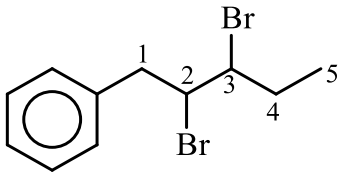
(NEW NCERT 12th Part-II Page No. 188)

85. (3)

Deficiency of vitamin B_2 causes cheilosis disease.

(NEW NCERT 12th Part-II Page No. 296)

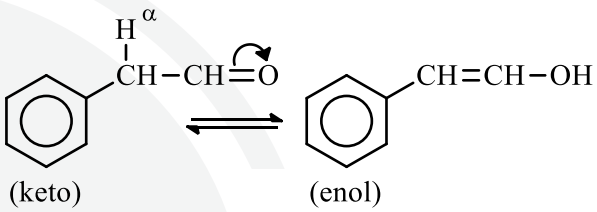
86. (1)



2,3-Dibromo-1-phenylpentane

(NEW NCERT 11th Part-II Page No. 270)

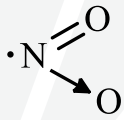
87. (1)



(NEW NCERT 12th Part-II Page No. 241)

88. (1)

NO_2 is an odd electron species.



(NEW NCERT 11th Part-I Page No. 105)

89. (2)

- $XeF_4 + O_2F_2 \xrightarrow{143\ K} XeF_6 + O_2$
- $2XeF_2 + 2H_2O \rightarrow 2Xe + 4HF + O_2$
- 1st compound of Xenon: $Xe^+PtF_6^-$
- Colourless explosive solid: XeO_3

(OLD NCERT 12th Part-I Page No. 204, 205)

90. (1)

For non-relativistic electron:

$$\lambda(\text{Å}) \approx \frac{12.27}{\sqrt{V(\text{eV})}}$$

$$\lambda \approx \frac{12.27}{12.27} = 1.00\ \text{Å}$$

(NCERT Ed. 2025-26 11th Part-I Page No. 49)

[Contd...

91. (4)

The given diagram is of smooth muscle fibres, which are tapered at both ends (fusiform), do not show striations, are bundled together in a connective tissue sheath, and are involuntary.

(NCERT Ed. 2020-21, 11th, Page No. 105)

92. (3)

Unicellular elongations of epidermal (parenchymatous) cells.

(NCERT Ed. 2025-26, 11th, Page No. 72)

93. (2)

The outer membrane does not form flattened sacs called thylakoids. The ribosomes of the chloroplasts (70S) are smaller than the cytoplasmic ribosomes (80S).

(NCERT Ed. 2025-26, 11th, Page No. 98)

94. (2)

A new strand always gets synthesised in 5' → 3' direction.

(NCERT Ed. 2025-26, 12th, Page No. 90)

95. (1)

By the end of 12 weeks (first trimester) of human pregnancy, most of the major organ systems of the foetus are formed, and structures like limbs and external genital organs become well developed. This stage marks significant early development of the human foetus.

(NCERT Ed. 2025-26, 12th, Page No. 38)

96. (1)

List-I	List-II
Leydig cells	Synthesize and secrete androgens
Sertoli cells	Provide nutrition to the male germ cells
First polar body	Formed after the first meiotic division of primary oocyte
Fimbriae	Finger-like projections that help in collection of the ovum

(NCERT Ed. 2025-26, 12th, Page No. 32)

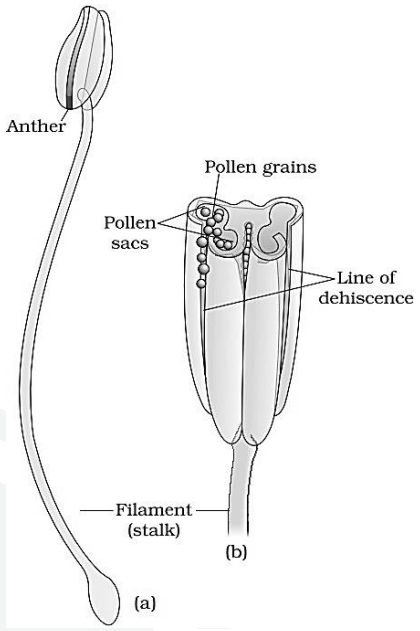
97. (4)

In the given diagram:

I-Anther

II-Filament (Stalk)

III-Pollen grains



(NCERT Ed. 2025-26, 12th, Page No. 5)

98. (4)

In oocytes of some vertebrates, diplotene can last for months or years.

(NCERT Ed. 2025-26, 11th, Page No. 126)

99. (2)

- Light reactions occur in the grana of chloroplasts.
- The LHC are made up of hundreds of pigment molecules bound to proteins.
- Photosystem II absorbs light at 680 nm.
- chlorophyll *b* absorb light and transfer the energy to chlorophyll *a*.

(NCERT Ed. 2025-26, 11th, Page No. 138)

100. (1)

- Taxonomic studies are useful for agriculture, forestry, and industry.
- Systematics helps classify organisms based on evolutionary relationships.

(NCERT Ed. 2025-26, 11th, Page No. 9)

101. (2)

The human heart develops from mesoderm and lies in the thoracic cavity between the lungs.

The SAN is located in the upper right corner of the right atrium, near the opening of the superior vena cava.

[Contd...

The AV bundle (Bundle of His) carries impulses from the AV node to the ventricles.
The pulmonary semilunar valve prevents backflow from the pulmonary artery into the right ventricle. The tricuspid valve prevents backflow from the right ventricle to the right atrium.

(NCERT Ed. 2025-26, 11th, Page No. 199)

102. (3)

Malignant tumors consist of proliferating neoplastic cells that grow rapidly, invade and damage surrounding tissues, starve normal cells by competing for vital nutrients, and exhibit metastasis (spreading to distant sites).

(NCERT Ed. 2025-26, 12th, Page No. 141)

103. (4)

In human blood, about 97% of oxygen is transported by red blood cells (RBCs) bound to haemoglobin.
The remaining ~3% of oxygen is transported dissolved in plasma.

(NCERT Ed. 2025-26, 11th, Page No. 189)

104. (4)

- tRNA brings amino acids and reads the genetic code.
- RNA polymerase binds to promoter and initiates transcription.

(NCERT Ed. 2025-26, 12th, Page No. 93)

105. (2)

List-I	List-II
P-wave	Depolarisation of atria
QRS complex	Depolarisation of ventricles
T-wave	Ventricles return to normal state (Repolarisation)
End of T-wave	End of ventricular systole

(NCERT Ed. 2025-26, 11th, Page No. 201)

106. (1)

The light bands contain actin and are called I-band (Isotropic), whereas the dark band is called 'A' (Anisotropic) band and contains myosin.

(NCERT Ed. 2025-26, 11th, Page No. 219)

107. (3)

Rapid population growth in India occurred mainly due to:
A decline in the death rate
Reduction in Maternal Mortality Rate (MMR)
Reduction in Infant Mortality Rate (IMR)
An increase in the number of people in the reproductive age group
These improvements in healthcare increased survival rates, leading to rapid population growth.

(NCERT Ed. 2025-26, 12th, Page No. 43)

108. (1)

Cerebellum has very convoluted surface in order to provide the additional space for many more neurons.

(NCERT Ed. 2025-26, 11th, Page No. 236)

109. (4)

Starch forms helical secondary structures and can hold I_2 molecules in the helical portion, producing a blue colour. Cellulose lacks complex helices and cannot hold I_2.

(NCERT Ed. 2025-26, 11th, Page No. 110)

110. (3)

List-I	List-II
Steroids	Cortisol, Testosterone
Iodothyronines	Thyroid hormones
IP_3	Secondary messenger
Proteins/Peptides	Pituitary hormones

(NCERT Ed. 2025-26, 12th, Page No. 248)

111. (1)

The descending limb of the loop of Henle is permeable to water but almost impermeable to electrolytes, leading to concentration of the filtrate.
The ascending limb of the loop of Henle is impermeable to water but allows electrolytes (Na^+, Cl^-) to be transported passively in the thin segment and actively in the thick segment.

(NCERT Ed. 2025-26, 11th, Page No. 209)

112. (4)

Delphinus (Common dolphin) is a mammal. Mammals are homoiothermous (warm-blooded), not cold-blooded.

(NCERT Ed. 2025-26, 11th, Page No. 51)

113. (3)

The thymus is quite large at the time of birth but keeps reducing in size with age, and by the time puberty is attained, it reduces to a very small size. It provides micro-environments for the maturation of T-lymphocytes.

(NCERT Ed. 2025-26, 11th, Page No. 243)

114. (3)

The 8th, 9th, and 10th pairs of ribs are called vertebrochondral (false) ribs because they join the seventh rib with hyaline cartilage. Only the 11th and 12th pairs are called floating ribs.

(NCERT Ed. 2025-26, 11th, Page No. 225)

[Contd...

115. (3)

Once the BOD of sewage or waste water is reduced significantly, the effluent is then passed into a settling tank where the bacterial 'flocs' are allowed to sediment. This sediment is called activated sludge. A small part of the activated sludge is pumped back into the aeration tank to serve as the inoculum. The remaining major part of the sludge is pumped into large tanks called anaerobic sludge digesters. Here, other kinds of bacteria, which grow anaerobically, digest the bacteria and the fungi in the sludge.

(NCERT Ed. 2025-26, 12th, Page No. 154)

116. (1)

Chlorella is used as a food supplement for space travel.Green algae usually have a rigid cell wall made of an inner layer of cellulose and an outer layer of pectose.

(NCERT Ed. 2025-26, 11th, Page No. 26)

117. (3)

Habitat loss and fragmentation	Cutting of Amazon rain forest for cultivation of soya beans
Over-exploitation	Steller's sea cow
Alien species invasions	The Nile perch introduced into Lake Victoria in east Africa
Co-extinctions	Extinction of plant leading to the extinction of its pollinator

(NCERT Ed. 2025-26, 12th, Page No. 222)

118. (3)

The thermostable DNA polymerase used in PCR is isolated from a bacterium called Thermus aquaticus. It remains active during the high temperature induced denaturation of double-stranded DNA.

(NCERT Ed. 2025-26, 12th, Page No. 173)

119. (2)

Polysomes consist of mRNA and ribosomes and are involved in protein synthesis.

(NCERT Ed. 2025-26, 11th, Page No. 91)

120. (1)

Among the animals, insects particularly bees are the dominant biotic pollination agents.

(NCERT Ed. 2025-26, 12th, Page No. 14)

121. (2)

The members of subphylum Vertebrata possess a notochord during the embryonic period, which is replaced by a vertebral column in adults.

Echinoderms does not contains vertebral column.

(NCERT Ed. 2025-26, 11th, Page No. 46)

122. (2)

Alfred Sturtevant	Mapped position of genes
Henking	X-body
Walter Sutton and Theodore Boveri	Chromosomal theory of inheritance
Thomas Hunt Morgan	Experimental verification of chromosomal theory of inheritance

(NCERT Ed. 2025-26, 12th, Page No. 67)

123. (3)

Sacred groves are found in Khasi and Jaintia Hills in Meghalaya, Aravalli Hills of Rajasthan, Western Ghat regions of Karnataka and Maharashtra and the Sarguja, Chanda and Bastar areas of Madhya Pradesh.

(NCERT Ed. 2025-26, 12th, Page No. 225)

124. (2)

Scientific evidence from astronomy and cosmology indicates that the universe is about 13.8 billion years old. This estimate comes from observations related to the Big Bang Theory, cosmic background radiation, and the expansion of galaxies.

(NCERT Ed. 2025-26, 12th, Page No. 111)

125. (1)

The gizzard (proventriculus) in cockroach has thick circular muscles and chitinous plates called teeth that mechanically grind the food into smaller particles before it enters the midgut for digestion.

(NCERT Ed. 2020-21, 11th, Page No. 113)

126. (3)

Hormones that interact with membrane-bound receptors (e.g., peptide hormones) do not enter the target cell. They bind to receptors on the cell surface and generate second messengers (like cAMP, IP_3) inside the cell.

Steroid hormones and iodothyronines are lipid-soluble, so they enter the target cell, bind to intracellular receptors, and regulate gene expression.

(NCERT Ed. 2025-26, 11th, Page No. 248)

[Contd...

127. (3)
ATP synthase is located in the inner mitochondrial membrane.
(NCERT Ed. 2025-26, 11th, Page No. 161)

128. (2)
Photorespiration occurs when RuBisCO binds with oxygen instead of carbon dioxide, leading to inefficient CO_2 fixation.
(NCERT Ed. 2025-26, 11th, Page No. 147)

129. (1)
Glomus help in absorption of phosphorus from soil by plants.
(NCERT Ed. 2025-26, 12th, Page No. 158)

130. (3)
Fatty acids enter the respiratory pathway after being converted to acetyl CoA.
(NCERT Ed. 2025-26, 11th, Page No. 162)

131. (4)
The quaternary structure is the architecture of a protein that is an assembly of more than one polypeptide or subunit (e.g., adult human haemoglobin with 4 subunits).
(NCERT Ed. 2025-26, 11th, Page No. 112)

132. (2)
MTPs are considered relatively safe during the first trimester, i.e., up to 12 weeks of pregnancy. Second-trimester abortions are much riskier.
(NCERT Ed. 2025-26, 12th, Page No. 46)

133. (3)
During normal (quiet) breathing, inspiration and expiration mainly involve the diaphragm and external intercostal muscles.
However, when strong or forced breathing is required (e.g., exercise, coughing, sneezing):
Forceful inspiration is helped by accessory muscles (like sternocleidomastoid and scalene muscles).
Forceful expiration is mainly aided by additional abdominal muscles, which push the diaphragm upward and reduce thoracic volume.
(NCERT Ed. 2025-26, 11th, Page No. 186)

134. (2)
AUG - 1st amino acid
UUU - 2nd amino acid
GUG - 3rd amino acid
UUU - 4th amino acid
UUC - 5th amino acid
UAA - Stop codon
Five amino acids will be coded by this mRNA sequence as UAA is stop codon and does not code for any amino acid.
(NCERT Ed. 2025-26, 12th, Page No. 96)

135. (2)
Ethylene promotes root hair formation, helping plant to increase absorption surface.
(NCERT Ed. 2025-26, 11th, Page No. 177)

136. (4)
ELISA is based on the principle of antigen-antibody interaction. It detects infections by the presence of antigens or by detecting the antibodies synthesized against the pathogen.
(NCERT Ed. 2025-26, 12th, Page No. 183)

137. (1)
- Bacterial viruses or bacteriophages (viruses that infect the bacteria) are usually double stranded DNA viruses.
- The viruses are non-cellular organisms that are characterised by having an inert crystalline structure outside the living cell.

(NCERT Ed. 2025-26, 11th, Page No. 20)

138. (2)
In dicot roots, the cortex is composed of thin-walled cells with many intercellular spaces.
(NCERT Ed. 2025-26, 11th, Page No. 74)

139. (1)

Chromatin	Stained by the basic dyes
Satellite	Non-staining secondary constrictions
Nucleolus	Site of active ribosomal RNA synthesis
Perinuclear space	Nuclear envelope

(NCERT Ed. 2025-26, 11th, Page No. 100)

140. (3)
In class Aves, the digestive tract has additional chambers, the crop and gizzard. Their skin is dry without glands except the oil gland at the base of the tail, their heart is completely four-chambered, and they are homoiothermous.
(NCERT Ed. 2025-26, 11th, Page No. 49)

141. (2)
Monocot roots lack secondary growth (and hence a vascular cambium), whereas many dicot stems develop a vascular cambium and undergo secondary thickening.
(NCERT Ed. 2025-26, 11th, Page No. 74)

[Contd...

142. (2)

The ventricles have much thicker muscular walls than the atria.

Ventricles must pump blood out of the heart—the right ventricle to the lungs and the left ventricle to the entire body—which requires generating high pressure.

(NCERT Ed. 2025-26, 11th, Page No. 199)

143. (4)

K (calyx) = 2+2

C (corolla) = 4

A (Androecium) = 2+4

G (Gynoecium) = 2

(NCERT Ed. 2025-26, 11th, Page No. 67)

144. (2)

The secondary structure of tRNA looks like a clover-leaf. In actual structure, the tRNA is a compact molecule which looks like inverted L.

(NCERT Ed. 2025-26, 12th, Page No. 98)

145. (1)

Volvox	Example of oogamous reproduction
Ulothrix	Example of isogamous reproduction with flagellated gametes
Spirogyra	Example of isogamous reproduction with non-flagellated gametes
Eudorina	Example of anisogamous reproduction

(NCERT Ed. 2025-26, 11th, Page No. 24)

146. (3)

In 1997, the first transgenic cow, Rosie, produced human protein-enriched milk (2.4 grams per litre). It contained human alpha-lactalbumin, which is a nutritionally more balanced product for human babies than natural cow-milk.

(NCERT Ed. 2025-26, 12th, Page No. 184)

147. (3)

The eye of the octopus and the eye of mammals are analogous structures. They are not anatomically similar but perform the same function, representing convergent evolution. Forelimbs and thorns/tendrils are examples of homology (divergent evolution).

(NCERT Ed. 2025-26, 12th, Page No. 115)

148. (1)

List-I	List-II
Trophoblast	Outer layer of blastocyst that gets attached to the endometrium
Inner cell mass	Contains certain cells called stem cells which have the potency to give rise to all tissues
Antrum	A fluid-filled cavity that characterizes a tertiary follicle
Zona pellucida	A new membrane formed by the secondary oocyte surrounding it

(NCERT Ed. 2025-26, 12th, Page No. 37)

149. (4)

- Tilman found that plots with more species showed less year-to-year variation in total biomass.
- Increased diversity contributed to higher productivity.

(NCERT Ed. 2025-26, 12th, Page No. 221)

150. (1)

Tidal Volume ≈ 500 mL. This is the normal volume of air inspired or expired during quiet breathing in a healthy adult.

Residual Volume is usually about 1100–1200 mL.

Inspiratory Reserve Volume averages 2500–3000 mL.

Vital Capacity is IRV + TV + ERV and does not include RV.

(NCERT Ed. 2025-26, 11th, Page No. 186)

151. (3)

Dragonflies help control the mosquito population.

(NCERT Ed. 2025-26, 12th, Page No. 157)

152. (3)

Glomerulonephritis refers specifically to inflammation of the glomeruli of the kidneys, which affects filtration.

(NCERT Ed. 2025-26, 11th, Page No. 214)

153. (4)

Compositae	Sunflower
Malvaceae	Cotton
Graminae	*Triticum*
Brassicaceae	Mustard
Malvaceae	Lady finger

(NCERT Ed. 2025-26, 11th, Page No. 68)

[Contd...

154. (1)

- The M phase is the shortest phase of the cell cycle.
- M phase is equational division.

(NCERT Ed. 2025-26, 11th, Page No. 121)

155. (3)

Natural classification systems developed, which were based on natural affinities among the organisms and consider, not only the external features, but also internal features, like ultrastructure, anatomy, embryology and phytochemistry.

(NCERT Ed. 2025-26, 11th, Page No. 24)

156. (3)

Parasympathetic neural signals (via the vagus nerve) act on the heart to decrease heart rate (negative chronotropic effect), reduce the speed of impulse conduction, especially through the AV node (negative dromotropic effect). This leads to a decrease in cardiac output.

(NCERT Ed. 2025-26, 11th, Page No. 202)

157. (3)

The axoplasm inside the axon contains a high concentration of K^+ and negatively charged proteins, and a low concentration of Na^+, whereas the fluid outside contains a low concentration of K^+ and a high concentration of Na^+.

(NCERT Ed. 2025-26, 11th, Page No. 232)

158. (4)

For multiplication of any foreign DNA in a host cell, it must be linked to an Origin of Replication (ori) so that the host cell can replicate it along with its own DNA.

The ori is the specific DNA sequence where replication begins, and it also helps regulate the copy number of the plasmid or linked DNA in the host cell.

(NCERT Ed. 2025-26, 12th, Page No. 164)

159. (2)

The main challenge for producing insulin via rDNA techniques was getting it assembled into a mature form. Eli Lilly solved this by producing chains A and B separately in *E. coli*, extracting them, and combining them by creating disulfide bonds.

(NCERT Ed. 2025-26, 12th, Page No. 182)

160. (1)

The adrenal cortex has three layers:

zona glomerulosa (outer), zona fasciculata (middle), and zona reticularis (inner).

Cortisol is a glucocorticoid, not a mineralocorticoid.

The main mineralocorticoid is aldosterone.

Aldosterone promotes Na^+ reabsorption (and hence water reabsorption) and K^+ excretion in kidney tubules.

Glucocorticoids actually stimulate gluconeogenesis and promote protein breakdown.

(NCERT Ed. 2025-26, 11th, Page No. 245)

161. (3)

- Males produce two different types of gametes, either with or without X-chromosome. Such type of sex determination mechanism is designated as male heterogamety. Ex.-Grasshopper, *Drosophila,* Human.
- Two different types of gametes in terms of the sex chromosomes, are produced by females in female Ex.-Birds.

(NCERT Ed. 2025-26, 12th, Page No. 71)

162. (4)

In frogs, the special venous connection between the intestine and liver is called the Hepatic portal system.

Blood from the intestine, rich in absorbed nutrients, first goes to the Liver through the portal vein.

The liver processes nutrients before the blood enters general circulation.

(NCERT Ed. 2025-26, 11th, Page No. 82)

163. (3)

Residual Volume (RV) is the air that remains in the lungs after forceful expiration. It is included only in Functional Residual Capacity.

Vital Capacity (VC) = IRV + TV + ERV

Inspiratory Capacity (IC) = TV + IRV

Functional Residual Capacity (FRC) = ERV + RV

Expiratory Capacity (EC) = TV + ERV

(NCERT Ed. 2025-26, 11th, Page No. 187)

164. (2)

Forest, grassland and deserts are natural terrestrial ecosystem while pond, wetland, estuary and lake are natural aquatic ecosystem. Crop fields (terrestrial ecosystem) and an aquarium (aquatic ecosystem) may also be considered as man-made ecosystems.

(NCERT Ed. 2025-26, 12th, Page No. 205)

[Contd...

165. (3)

- Majority of the parasites harm the host.
- Many marine fish are infested with ectoparasitic copepods.

(NCERT Ed. 2025-26, 12th, Page No. 200)

166. (2)

According to the theory of special creation described in conventional religious literature, Earth was believed to be about 4000 years old, and all organisms were created in their present form at the time of creation.

(NCERT Ed. 2025-26, 12th, Page No. 112)

167. (2)

Heroin, commonly called smack, is chemically diacetylmorphine. It is obtained by the acetylation of morphine, which is extracted from the latex of the poppy plant *Papaver somniferum*.

(NCERT Ed. 2025-26, 12th, Page No. 142)

168. (4)

Oxygen is not directly used in the TCA cycle, but it is involved in the electron transport chain.

(NCERT Ed. 2025-26, 11th, Page No. 159)

169. (2)

Hind II always cuts DNA molecules at a particular point by recognizing a specific sequence of six base pairs, known as its recognition sequence.

(NCERT Ed. 2025-26, 12th, Page No. 165)

170. (1)

In cortical nephrons, the vasa recta is either absent or poorly developed.

The loop of Henle in cortical nephrons is very short and extends only slightly into the medulla.

(NCERT Ed. 2025-26, 11th, Page No. 208)

171. (3)

Amongst plants, the highest species diversity is shown by angiosperms (flowering plants), followed by algae, mosses (bryophytes) and ferns (pteridophytes).

(NCERT Ed. 2025-26, 12th, Page No. 222)

172. (2)

Sutures are fibrous joints.

Between vertebral bodies there are cartilaginous joints (symphysis) with intervertebral discs.

The atlanto-axial joint is a pivot type synovial joint.

The carpometacarpal joint of the thumb is a saddle synovial joint.

The shoulder joint is a ball-and-socket synovial joint.

(NCERT Ed. 2025-26, 11th, Page No. 227)

173. (3)

- Humification leads to accumulation of a dark coloured amorphous substance called humus that is highly resistant to microbial action and undergoes decomposition at an extremely slow rate.
- The rate of biomass production is called productivity.

(NCERT Ed. 2025-26, 12th, Page No. 207)

174. (3)

Sewage water is treated to reduce BOD before being released into natural water bodies. The greater the BOD of waste water, more is its polluting potential.

(NCERT Ed. 2025-26, 12th, Page No. 154)

175. (3)

The process in which a plant's cells lose the ability to divide and specialize into specific functions is called differentiation.

(NCERT Ed. 2025-26, 11th, Page No. 172)

176. (3)

- Statement A is correct. The Earth is currently experiencing the sixth mass extinction, mainly driven by human activities such as habitat loss, overexploitation, climate change, and pollution.
- Statement B is incorrect. Greater solar energy and higher productivity occur in the tropical regions, not temperate regions, which explains higher biodiversity near the equator.
- Statement C is correct. The species–area relationship was first explained by Alexander von Humboldt, showing that species richness increases with area.
- Statement D is incorrect. The Western Ghats have much higher amphibian species diversity than the Eastern Ghats and are a major biodiversity hotspot.

Hence, the correct set of statements is A and C only (Option C).

(NCERT Ed. 2025-26, 12th, Page No. 220)

177. (4)

If a child's blood group is 'O'. The genotype of mother and father blood groups can be heterozygous A and heterozygous A.

(NCERT Ed. 2025-26, 12th, Page No. 61)

[Contd...

178. (3)

India is one of the world's mega-diverse countries. About 45,000 plant species and nearly 90,000 animal species (roughly twice the number of plants) have been recorded from India.

- Option A is incorrect because India has about 1,200–1,300 bird species, not more than 1,500.
- Option B is incorrect as India's share of global species diversity is about 8.1%, not 18.1%.
- Option D is incorrect because India occupies only ~2.4% of the world's land area, not 4.2%.

(NCERT Ed. 2025-26, 12th, Page No. 219)

179. (2)

The first stage is the protonema stage, which develops directly from a spore. It is a creeping, green, branched and frequently filamentous stage. The second stage is the leafy stage, which develops from the secondary protonema as a lateral bud. The leafy stage develops from the secondary protonema and bears the sex organs for sexual reproduction.

(NCERT Ed. 2025-26, 11th, Page No. 30)

180. (3)

A nucleotide has three chemically distinct components: a heterocyclic compound (nitrogenous base), a monosaccharide (pentose - ribose or deoxyribose), and a phosphoric acid or phosphate.

(NCERT Ed. 2025-26, 11th, Page No. 111)

■■■